The U.S. Clinical-Trial Investigator Supply

A preliminary descriptive analysis of FDA BMIS Form 1572 filings, benchmarked against industry trials; China CFDI registry supplement

Strict persons · 2009–2025 · generated 2026-11-07

PRELIMINARY ANALYSIS ONLY — NOT FOR DECISION-MAKING

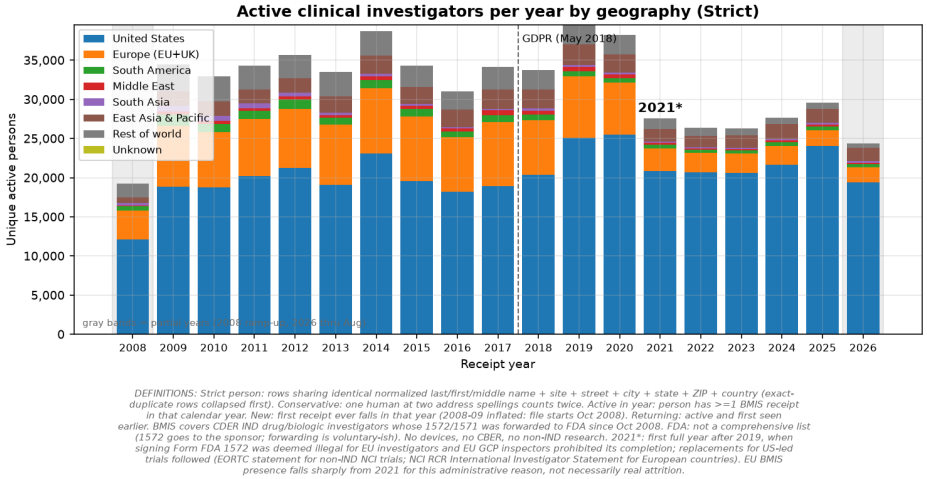
FDA BMIS (Form 1572, thru Aug 2026) · clinicaltrials.gov API v2 · github.com/henryweimd/bmis1572

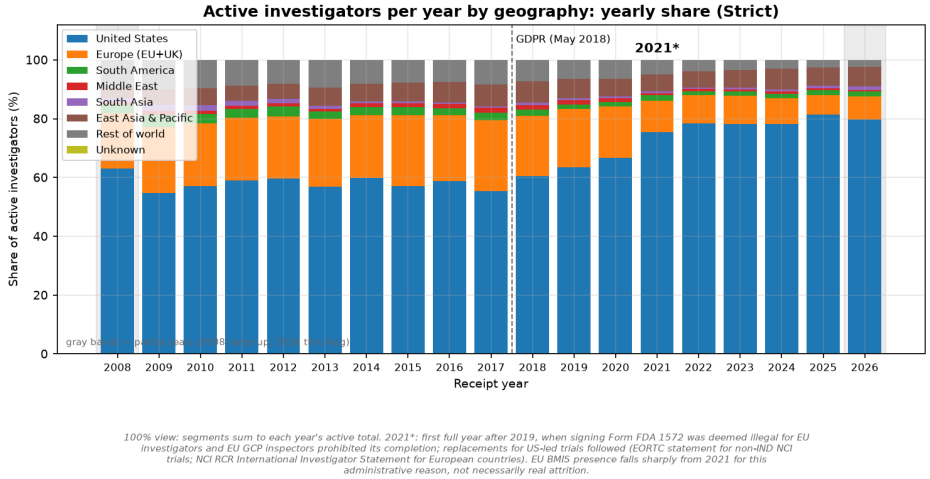
What this is (and isn’t)

- Supply = clinical investigators (CIs) named on Form 1572 filings in FDA’s BMIS file.
- Persons are name+site clusters: Strict (headline), Merged, Raw bound the identity call.
- Active in year = ≥1 receipt; headline window 2009–2025.
- Demand = industry-led interventional trials (clinicaltrials.gov API), starts + running.
- Benchmarks are year-level ratios — no record-level CI↔trial linkage exists.

BMIS covers CDER IND drugs/biologics only; FDA states it is not comprehensive.
2008 is file ramp-up, 2026 partial (thru Aug); both excluded from growth stats.

Headline: global flat-to-down, U.S. up

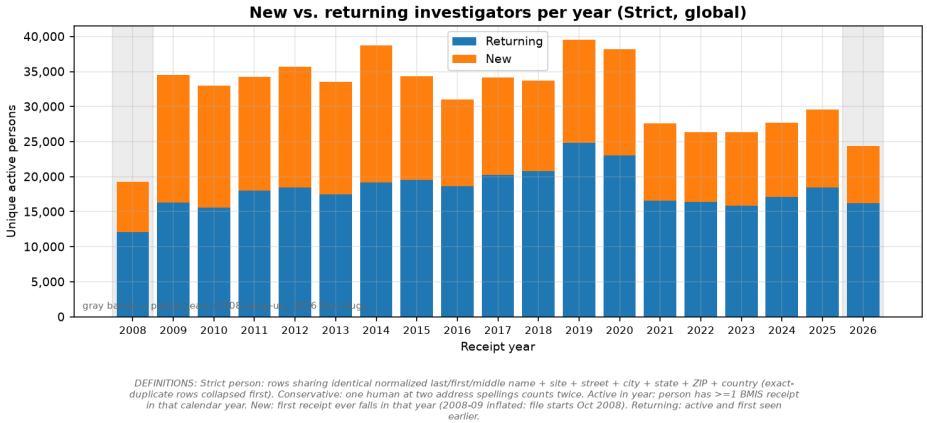


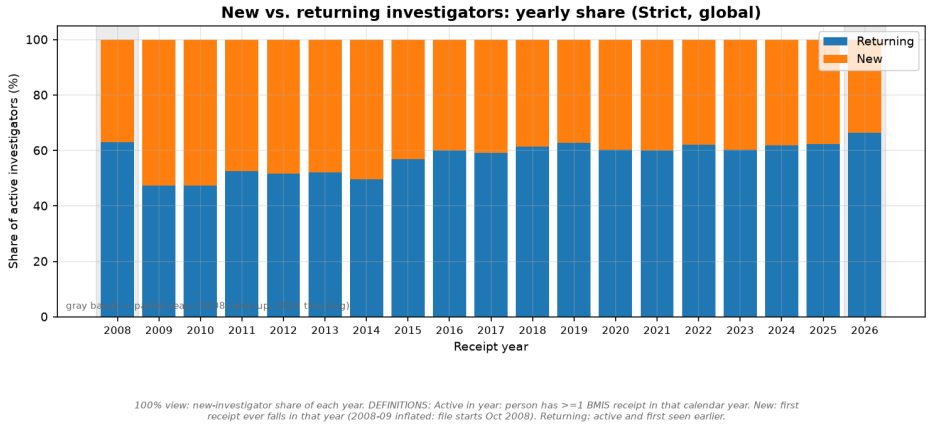


- Global active CIs: 34,468 → 29,596 (-14.1%).
- U.S. active CIs: 18,838 → 24,084 (+27.8%).
- Ex-U.S. decline concentrates in 2020–21; no EU break at GDPR (May 2018).
- 2021* follows the 2019 EU Form 1572 restriction (EORTC + NCI RCR replacements): the EU drop is administrative, not proven attrition.

Strict persons; receipt-year activity. Right = yearly share: EU 19.9% (2019) → 10.7% (2021).

Fewer newcomers; a stable returning core

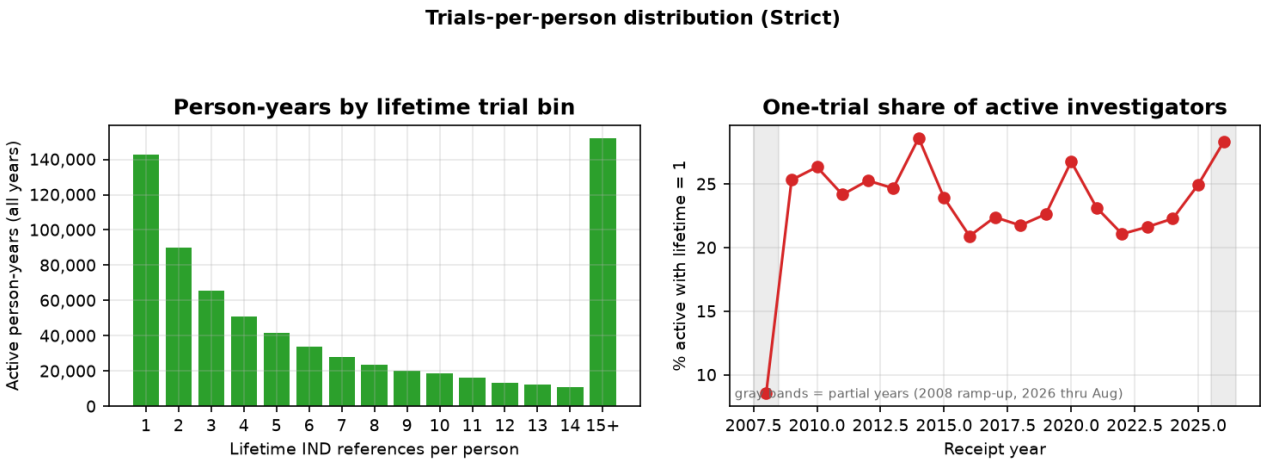




- 38% of 2025 actives are first-timers; the rest return.
- Entry is dominated by one-and-done dynamics (next).
- Right panel: new-investigator share of each year.

New = first receipt ever in that year.

Half of investigators are one-and-done

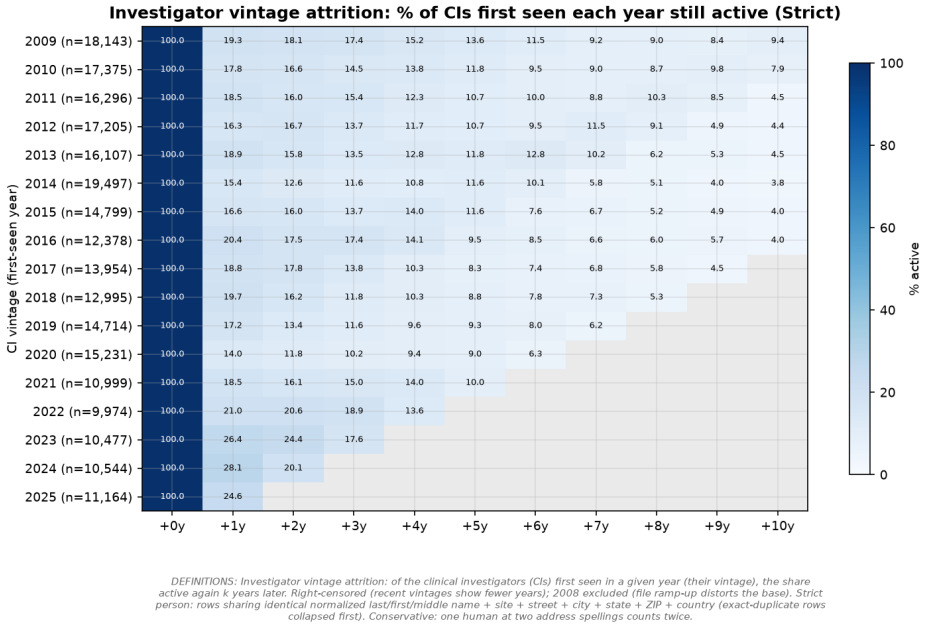


DEFINITIONS: Trials per person: collapsed BMIS rows per person-cluster. One row = one IND submission reference, a proxy for trials, not confirmed distinct trials. One-trial (one-and-done): person-cluster linked to exactly one IND reference across the full 1980-2026 file span.

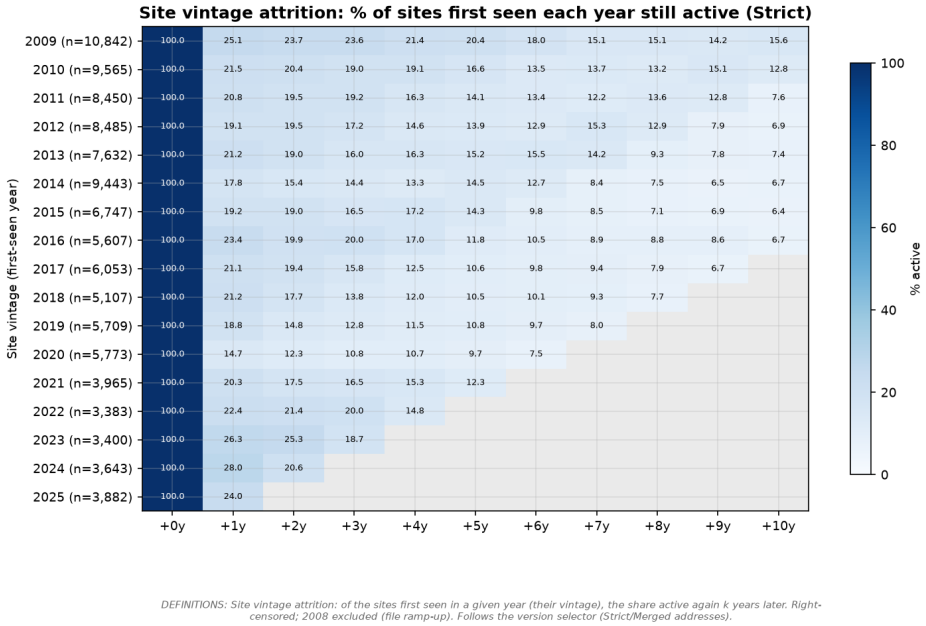
- 49.3% of persons have exactly one lifetime IND reference.
- Median 2, mean 3.85 trials per investigator.
- A thin right tail of hyper-active CIs carries the top bins.

One row = one IND reference: a proxy, not confirmed distinct trials.

Vintage attrition: steep, then flat



DEFINITIONS: Investigator vintage attrition: of the clinical investigators (CIs) first seen in a given year (their vintage), the share active again k years later. Right-censored (recent vintages show fewer years); 2008 excluded (file ramp-up distorts the base). Strict person: rows sharing identical normalized last/first/middle name + site + street + city + state + ZIP + country (exact-duplicate rows collapsed first). Conservative: one human at two address spellings counts twice.

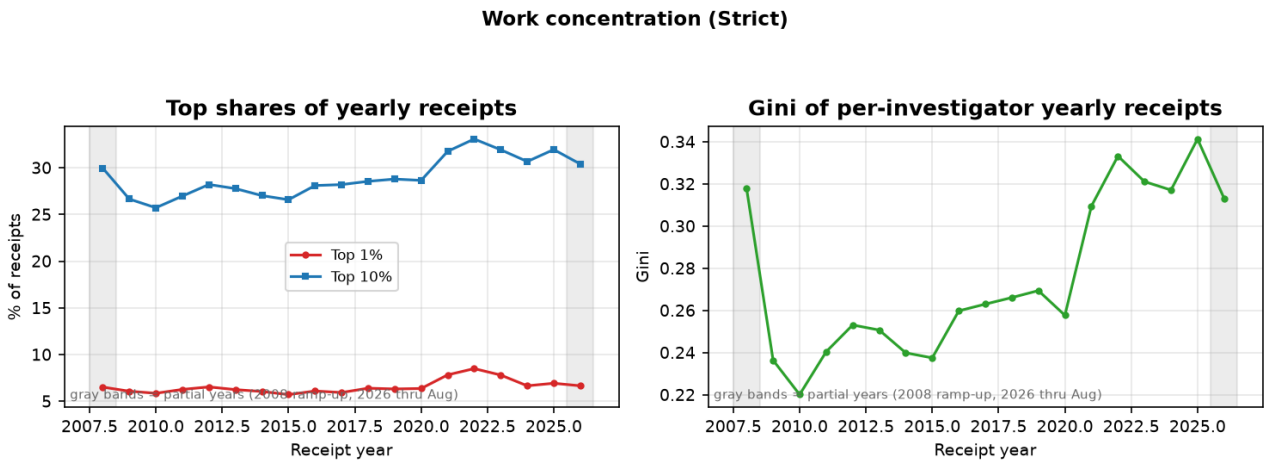


DEFINITIONS: Site vintage attrition: of the sites first seen in a given year (their vintage), the share active again k years later. Right-censored; 2008 excluded (file ramp-up). Follows the version selector (Strict/Merged addresses).

- ≈19% of the 2009 CI vintage is active a year later; ≈9% a decade later.
- Curves look alike across vintages: most attrition is immediate.
- Sites persist longer: ≈25% / ≈16% for the 2009 site vintage.

Vintage = first seen in that year. Right-censored; 2008 excluded.

Work is concentrating

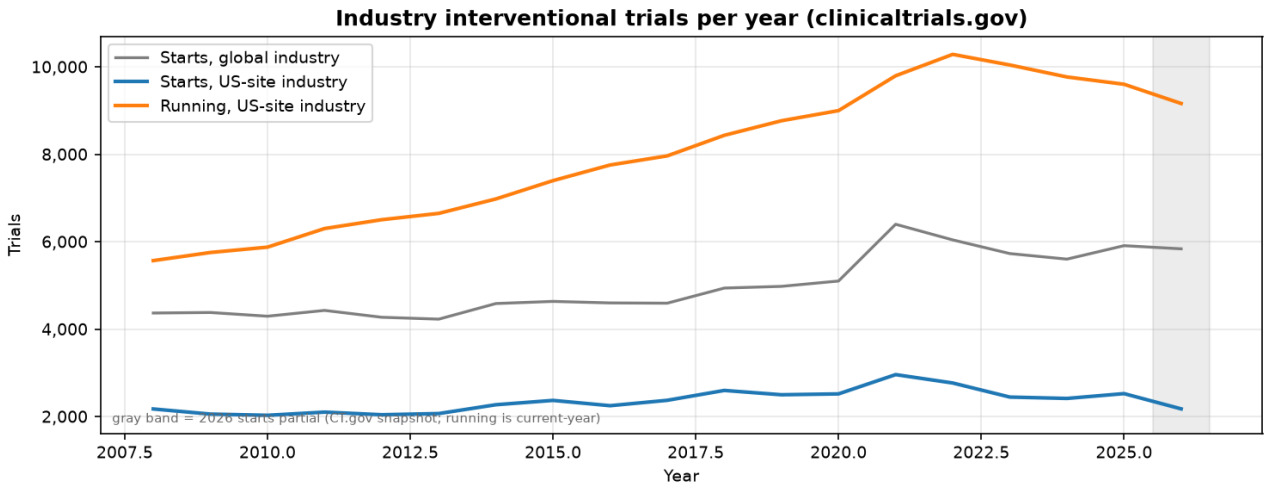


DEFINITIONS: Annual concentration: share of each year's receipts going to the year's top 1%/10% investigators by receipts, plus the Gini of per-investigator yearly receipts. Higher means work concentrated in fewer hands.

- Top-10% receipt share: 27% → 32%.
- Gini of yearly receipts: 0.24 → 0.34.

Annual within-year concentration among actives.

Demand: industry trial starts

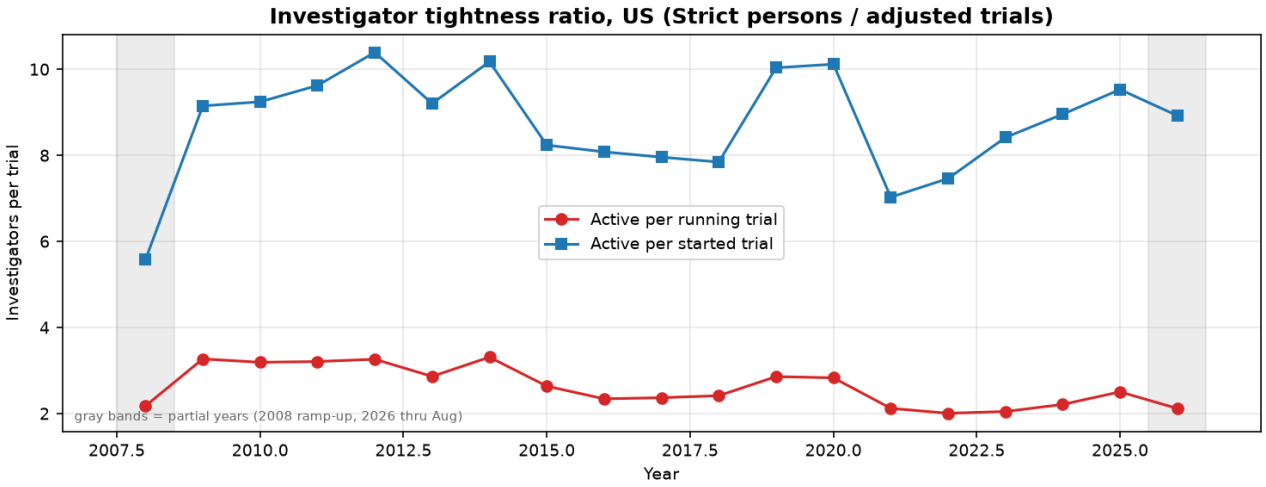


DEFINITIONS: Unadjusted benchmark: all TYPE=CI BMIS persons vs. all global industry-led interventional trial starts (clinicaltrials.gov). Static charts and the explorer Tightness scope toggle. Adjusted benchmark: US-band BMIS persons vs. industry-led interventional trials with >=1 US site. Default scope of the explorer's Tightness chart. Starts: trials by Study Start Date year. Running: start <= year-end AND (completion >= year-start OR still active with no completion date). CT.gov snapshot date recorded in manifest.

- Global industry starts: 5,911 in 2025.
- U.S.-site running trials: 9,604 in 2025.
- Starts and supply are not independent — ratios describe, not explain.

clinicaltrials.gov API v2, interventional + industry lead sponsor.

Tightness: investigators per running trial

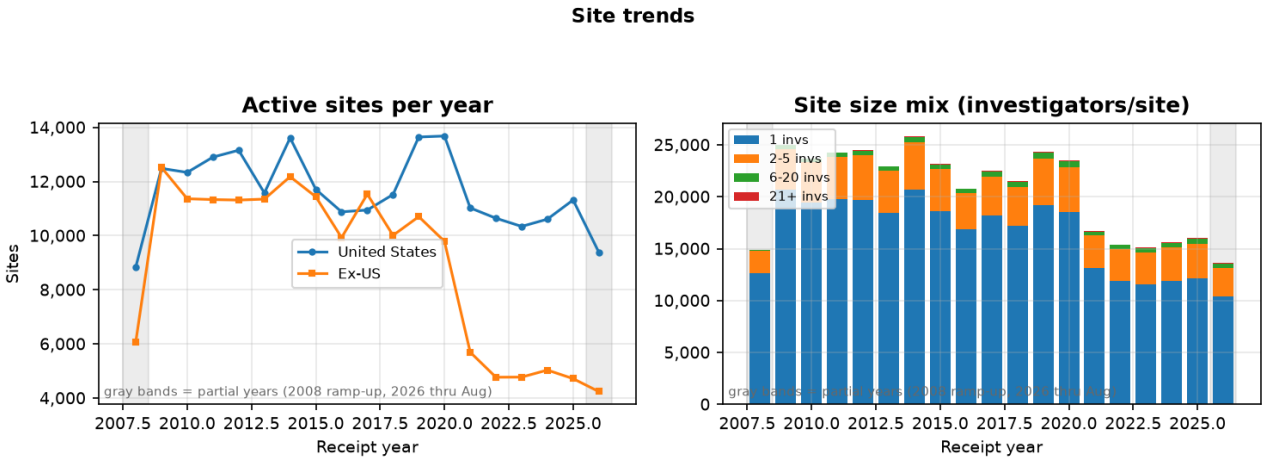


DEFINITIONS: Tightness ratio: active investigators per running trial. Descriptive workload proxy only; supply and trial counts are not independent (a supply crunch can suppress starts), so this is not causal. Active in year: person has >=1 BMIS receipt in that calendar year. New: first receipt ever falls in that year (2008-09 inflated: file starts Oct 2008). Returning: active and first seen earlier. Adjusted benchmark: US-band BMIS persons vs. industry-led interventional trials with >=1 US site. Default scope of the explorer's Tightness chart.

- U.S. CIs per running trial: 3.3 → 2.5.
- Adjusted scope: U.S. investigators vs. U.S.-site trials.

Ecological ratio; supply constraints can move trial counts.

Sites: where the work happens

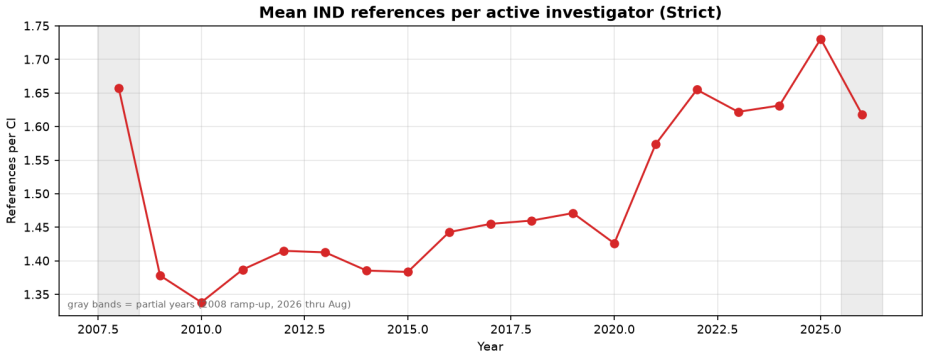


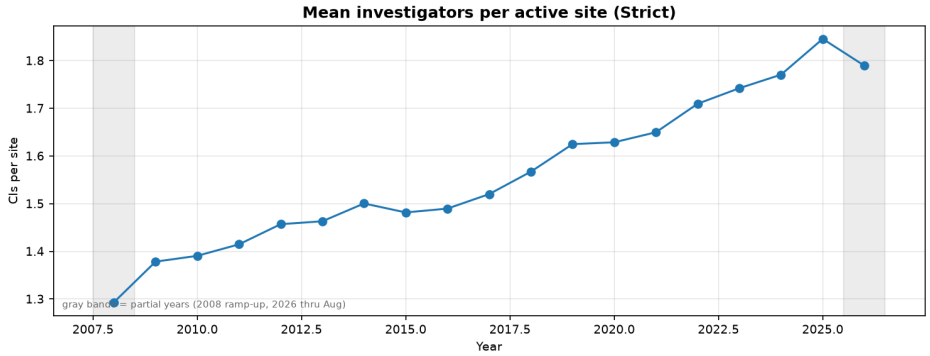
DEFINITIONS: A site is a PLACE (street address): Strict matches normalized street+city+state+country exactly (suite/floor markers dropped, ZIP excluded); Merged additionally unions same-house-number addresses with street-token Jaccard >= 0.6 within a city. Branches of one org at different addresses stay split. Site rollups use strict persons. 2008 (file ramp-up from Oct 1) and 2026 (file through Aug 31) are partial: shown faded, excluded from growth statistics.

- 16,036 active sites in 2025 (Strict addresses).
- Solo-investigator sites dominate; mega-centers are few.
- All-time leader: THE UNIVERSITY OF TEXAS MD ANDERSON CANCER CENTER (847 CIs, 5,390 refs).

Site = street address; branches of one org stay split.

Intensification: fewer hands, more load



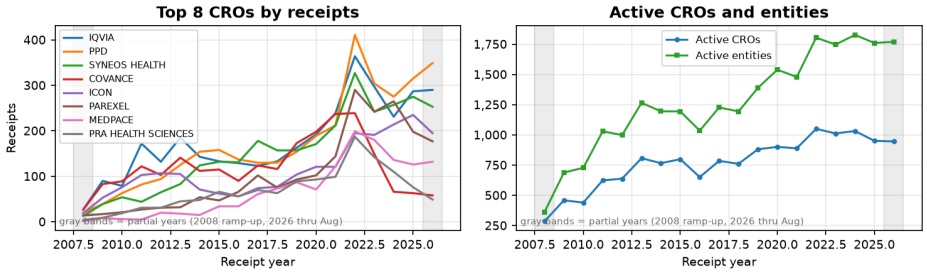


- References per active CI: 1.38 → 1.73.
- CIs per active site: 1.38 → 1.85.
- Work concentrates on repeat investigators at busier sites.

Strict persons/addresses; 2008 and 2026 partial.

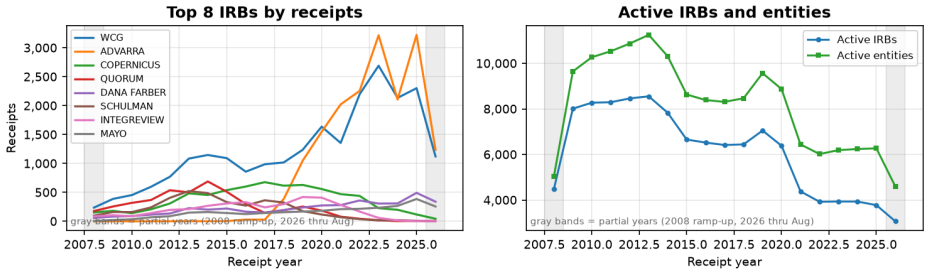
CROs and IRBs in context

CRO trends



DEFINITIONS: CRO views group BMIS TYPE=CRO rows by normalized company name: evidence-based aliases for known giants (PPD, Medpace, IQVIA, PRA, Syneos, ICON, Parexel, Covance, Celerion, others), else first site segment with suffixes/punctuation stripped. Same-name companies materially spanning states are state-qualified; aliases stay global. Residual variants stay split; cro_index lists variants. 2008 (file ramp-up from Oct 1) and 2026 (file through Aug 31) are partial: shown faded, excluded from growth statistics.

IRB trends



DEFINITIONS: IRB views group BMIS TYPE=IRB rows by normalized board name: evidence-based aliases for known boards (Advarra, WCG/Western, Copernicus, IntegReview, Schulman, Quorum, Sterling, Chesapeake), else first non-empty site segment with IRB stopwords, room markers, and numbers stripped. Same-name boards materially spanning states are state-qualified; aliases stay global. Residual variants stay split; irb_index lists variants. 2008 (file ramp-up from Oct 1) and 2026 (file through Aug 31) are partial: shown faded, excluded from growth statistics.

- 951 CROs and 3,779 IRBs active in 2025.
- All-time receipt leaders: PPD / WCG.

Normalized company/board groupings; residual variants audited.

Demand side: US industry trials

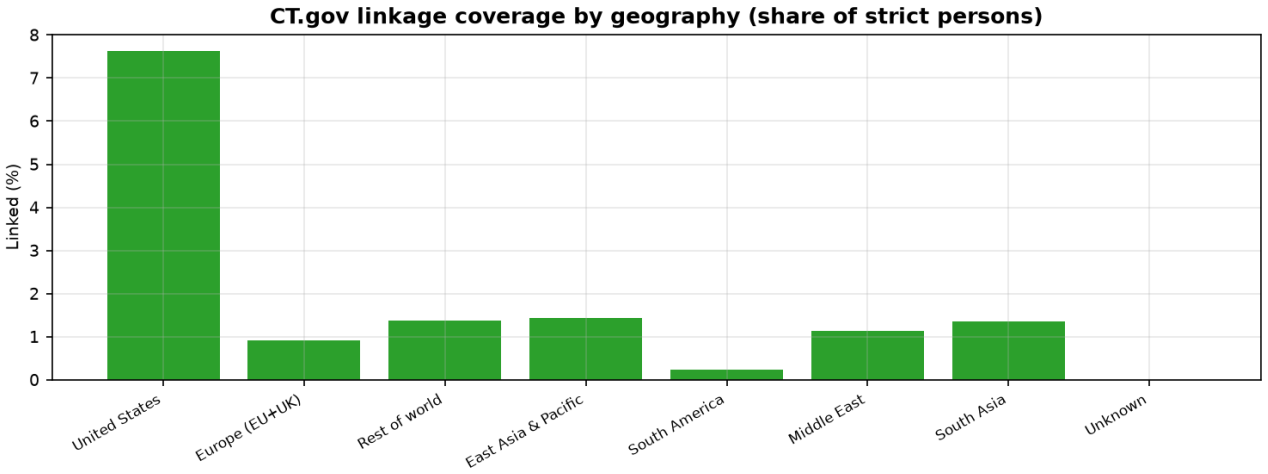


Exact per-trial lead sponsor; variants merged by normalized key. US industry-led interventional trials, all years in file.

- 53,543 US industry-led interventional trials from 7,940 sponsors.
- Top sponsor: Pfizer (1,760 trials).
- Top phase: P1 (12,914 trials).
- 38,731 trials (72%) name ≥1 study official.

clinicaltrials.gov API v2, all years in file.

Person linkage: BMIS ↔ CT.gov (experimental)

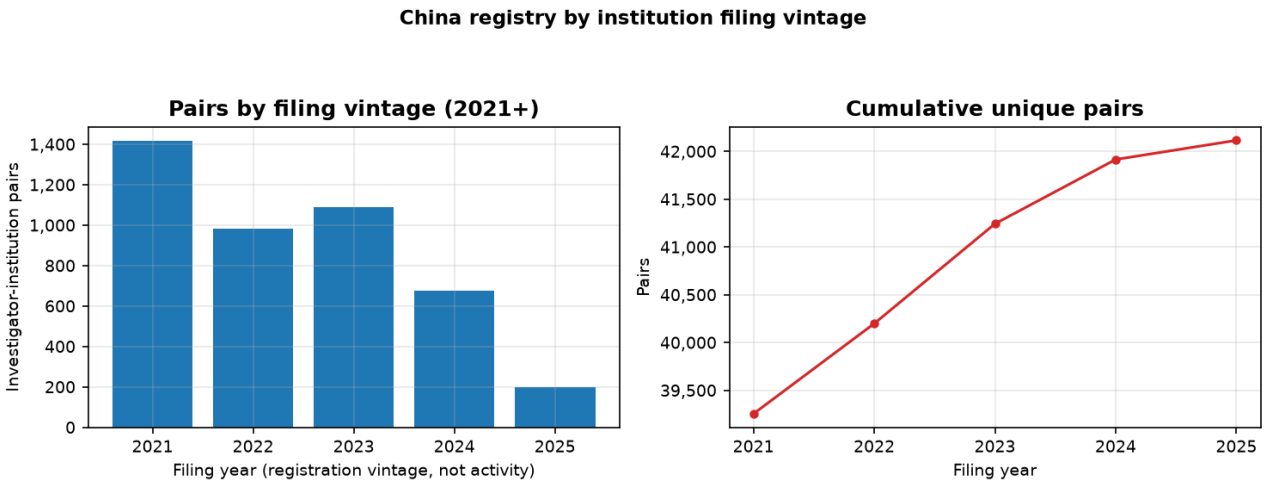


DEFINITIONS: CT.gov person linkage (experimental): BMIS strict persons matched to overall officials (usually study-level directors/chairs; median 1 per trial, though some trials list site PIs) by normalized name, gated on affiliation-to-site token overlap. Strong = full-name match + >=2 overlap tokens; weak = full-name + 1 token (or no affiliation) or initial + >=2 tokens. Covers ~4% of persons; weak links may be collisions.

- 13,956 BMIS persons match ≥1 study official (11,961 strong / 12,696 weak links).
- Most-linked: Williams, Dorian (43 trials).
- Name + affiliation-overlap match; most trials name only study-level officials, so coverage is thin and KOL-skewed.

Weak links may be name collisions. See Definitions.

China: registry stock by filing vintage

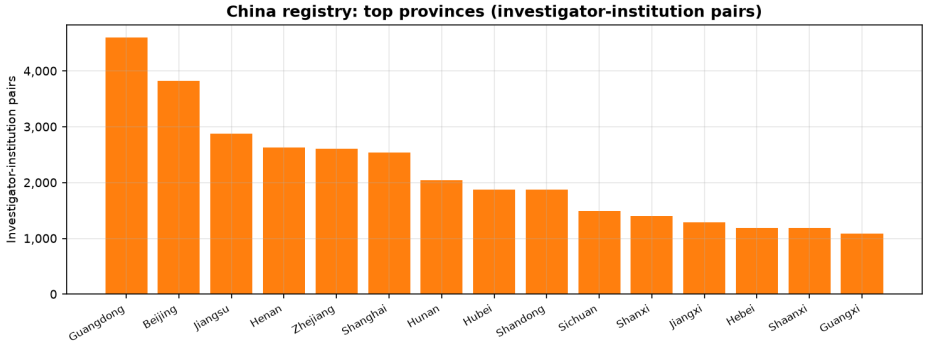


DEFINITIONS: China CFDI registry extract (partial coverage, retrieved 2026-11-06): investigators affiliated with filed drug-trial institutions. Person = (name, institution) pair; rows exact-deduped at ingest (the extract repeats ~35% of rows). Pairs count affiliations, not humans: one person at two institutions = two pairs; same-named people at one institution = one pair; common names (e.g. 王伟) collide across institutions and are not mobility. Filing year is the INSTITUTION's registration vintage, not investigator activity: most institutions filed in a 2019-2020 bulk wave. YoY reads as registry composition, never retention. 2020 bulk-registration spike (33,476 pairs) omitted from bars; cumulative counts each pair once at its first vintage.

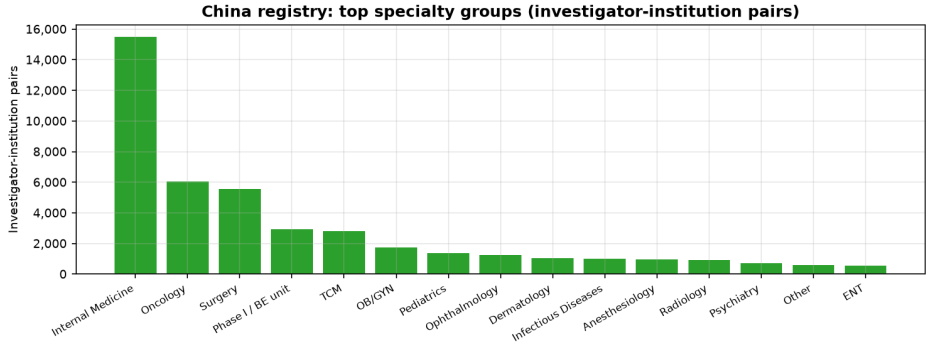
- 42,117 investigator-institution pairs; 34,077 names; 1,818 institutions.
- Bulk 2019–20 registration wave (33,476 pairs in 2020); graphs read 2021+.
- Filing year is institution vintage, not investigator activity.

Pairs count affiliations, not humans; common names collide. Partial extract, retrieved 2026-11-06.

China: provinces & specialties



DEFINITIONS: China CFDI registry extract (partial coverage, retrieved 2026-11-06): investigators affiliated with filed drug-trial institutions. Person = (name, institution) pair; rows exact-deduped at ingest (the extract repeats ~35% of rows). Pairs count affiliations, not humans: one person at two institutions = two pairs; same-named people at one institution = one pair; common names (e.g. □□) collide across institutions and are not mobility. ZH is authoritative. EN covers provinces, cities, roles, specialty groups, top-30 sites (curated) and pattern-derived site/specialty labels; investigator names stay in Chinese.

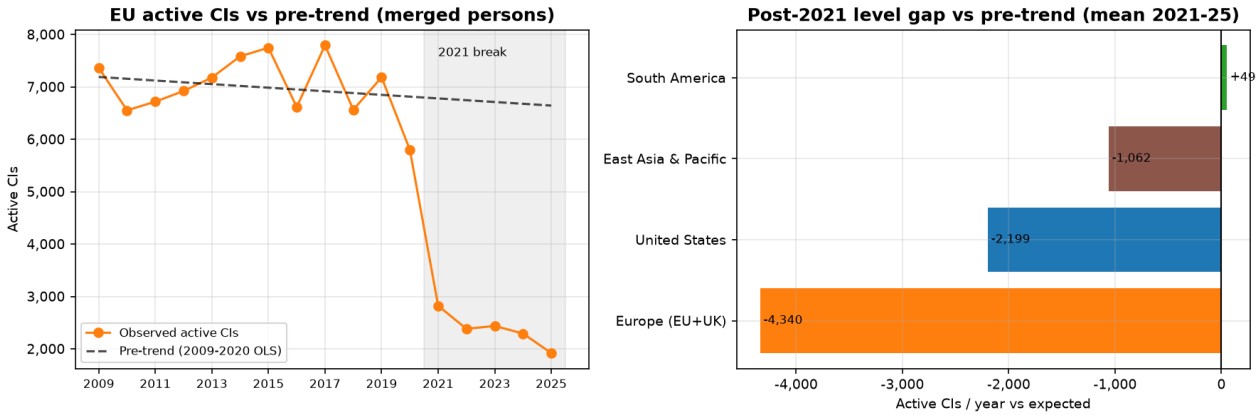


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- Top province: Guangdong (4,600 pairs).
- Top specialty group: Internal Medicine (15,515 pairs).

ZH authoritative; EN labels curated or pattern-derived.

EU break: permanent, EU-specific — descriptive

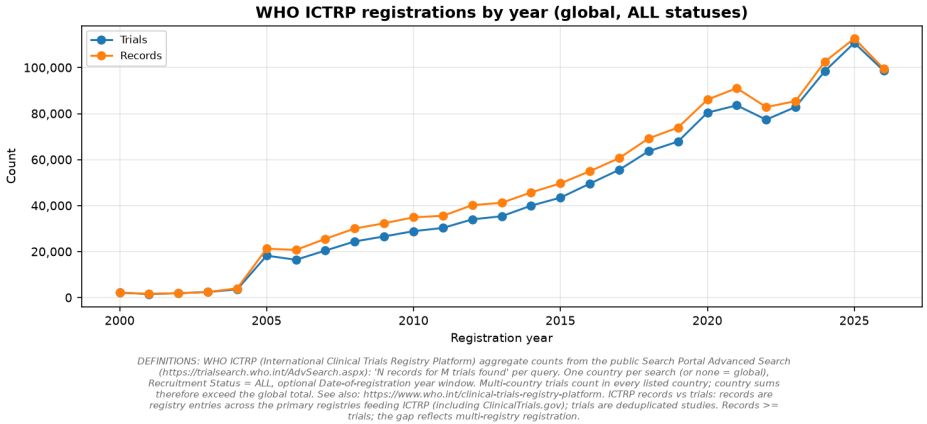


EU 2021 gap -4,340 CIs/yr (-62% of pre-mean); placebo EU gaps -1,148 (2016) and -506 (2017). Per pre-registered spec (docs/eu_break.md) the study ships descriptive-only: the US share 'surge' is the compositional mirror of the EU exit and EU mid-sample volatility trips the placebo gate. Timing coincides with the 1572 prohibition + 2021 NCI RCR replacement.

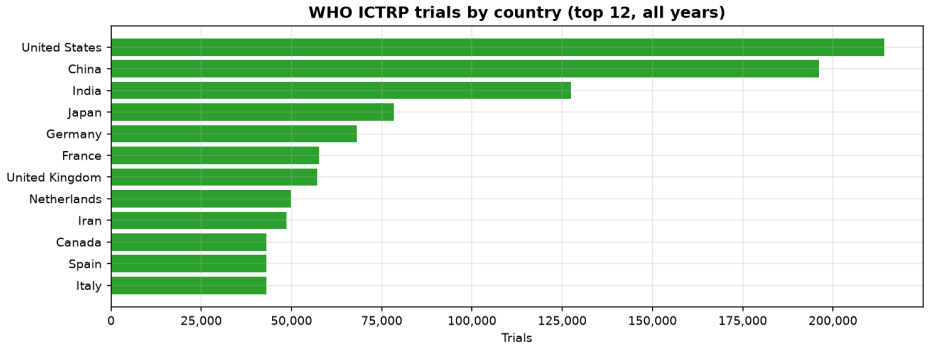
- EU runs 4,339/yr below its 2009–20 pre-trend (62% of pre-mean); share gap -10.7pp.
- Placebo EU gaps (-1,148, -505) are 3–8× smaller; no control band collapses in level terms.
- US share ‘surge’ is the compositional mirror of the EU exit (US level fell too) — the denominator collapsed.
- Pre-registered verdict: descriptive-only. Timing coincides with the 1572 prohibition + 2021 NCI RCR replacement.

Merged persons; pre-trend OLS 2009–20, post mean 2021–25. Spec: docs/eu_break.md.

Global context: WHO ICTRP



DEFINITIONS: WHO ICTRP (International Clinical Trials Registry Platform) aggregate counts from the public Search Portal Advanced Search (https://trialsearch.who.int/AdvSearch.aspx): 'N records for M trials found' per query. One country per search (or none = global), Recruitment Status = ALL, optional Date-of-registration year window. Multi-country trials count in every listed country; country sums therefore exceed the global total. See also: https://www.who.int/clinical-trials-registry-platform. ICTRP records vs trials: records are registry entries across the primary registries feeding ICTRP (including ClinicalTrials.gov); trials are deduplicated studies. Records >= trials; the gap reflects multi-registry registration.



Multi-country trials repeat in every listed country; do not sum. DEFINITIONS: ICTRP records vs trials: records are registry entries across the primary registries feeding ICTRP (including ClinicalTrials.gov); trials are deduplicated studies. Records >= trials; the gap reflects multi-registry registration.

- 1,175,909 trials / 1,285,607 records worldwide.
- Top: United States (214,203), China (196,046), India (127,479).
- Multi-country trials repeat per country; registration year, ALL statuses.

Aggregate portal counts retrieved 2026-11-07; no trial-level data.

Limitations & what’s next

- No therapeutic-area or phase fields in BMIS; rows carry date + site only.
- Person↔trial linkage is name-based, thin and KOL-skewed; site cross-check is institution×city (both experimental).
- Person/site counts are ranges across versions, not exact headcounts.
- Next: therapeutic-area mix via matched trials; global linkage; cross-site tracking.

Sources & reproducibility

- Supply: FDA BMIS quarterly file — https://www.fda.gov/drugs/drug-approvals-and-databases/bioresearch-monitoring-information-system-bmis
- Demand: clinicaltrials.gov Data API v2 — https://clinicaltrials.gov/data-api/api
- Global context: WHO ICTRP Search Portal — https://trialsearch.who.int/AdvSearch.aspx
- China: CFDI drug-institution filing extract — https://beian.cfdi.org.cn/CTMDS/apps/pub/drugPublic1.jsp
- All code, methods, definitions: https://github.com/henryweimd/bmis1572
- Rebuild locally: uv run python pipeline/build.py